

Practice Task 2

MS Excel Function

Description: We did every function of the excel based on the given task, we just have to make it work every function to work.

Screenshots of the tasks:

1.Logical Function

IF

What does it do?

It checks whether a condition is true or not and on the basis of that returns a value

Syntax

`=if(Logical_Test, [value_if_true], [value_if_false])`

Logical_Test

This will be the condition which you want to check, so greater than, less than or equal to signs can be used for numbers. When you want to evaluate a condition based on **text value**, then use double quotes (") with equal to signs (Row 27 for example)

Value_if_True

If the logical test holds, then excel will return this value

Value_if_False

If the logical test does not hold, then excel will return this value

Example

Using IF formula to identify if a student has passed an exam or not

Student ID	Marks	Pass or Fail	Formula
S001	90	Pass	=IF(B22>=40,"Pass","Fail")
S002	70	Pass	=IF(B23>=40,"Pass","Fail")
S003	20	Fail	=IF(B24>=40,"Pass","Fail")
S004	41	Pass	=IF(B25>=40,"Pass","Fail")

Note

If instead of marks, we had Grades A, B, C i.e. **text values** then the above formulas would have changed by
IF(B21="A","Pass","Fail")

Exercise

Write a formula using IF. Salary for an AVP will be \$2,000 and for an analyst will be \$1,000

Emp ID	Designation	Salary
E001	AVP	\$2000
E002	Analyst	\$1000
E003	Analyst	\$1000
E004	Analyst	\$1000

< > **Logical - IF** Logical - AND Logical - Nested IFs Logical - OR Logical - IFERROR Logical - IFNA T

2. Text Function

EXACT

What does it do?
It checks whether 2 values (texts, numbers) are exactly same or not. If they are same, then it will return TRUE else it will return FALSE. This function is case-sensitive

Syntax
=EXACT(Text1, Text2)

Text1
First string which you want to check

Text2
Second string which you want to test against

Example 1
Testing condition 1 and condition 2 to check the numbers are greater than 0 using AND

Col A	Col B	Result	Formula
EXCEL	EXCEL	TRUE	=EXACT(A19,B19)
EXCEL	Excel	FALSE	=EXACT(A20,B20)

Note
In the above example, Row 20 result is FALSE because the EXACT function is case-sensitive. The first value is in upper capse [EXCEL] whereas the second one (Excel) is a combination of upper and lower capse

Exercise
Write a formula using EXACT to check if the values in Col A and Col B are same or not

Col A	Col B	Results
TEST	Test	TRUE
TEST	TEST	TRUE
1234	1234	TRUE
1.1	1.101	FALSE

Text - EXACT | Text - TEXT | Text - CONCATENATE | Text - TRIM | Text - SUBSTITUTE | Text - VALUE | Text - UPPER | Text - LOWER | Text - LEN | Text - LEFT | Text - RIGHT | Text - MID

3. Date Function

DATE

What does it do?
This function returns the date when days, month, year are stored in different cells. If the cell in which the function is being called is of Number format, then the function will return a number - excel stores dates as numbers

Syntax
=DATE(Year,Month,Day)

Year
Year of the date. Try to supply the entire 4 digits

Month
Month of the date in numeric format like for January it would be 1, for February it would be 2

Day
Day of the date - it needs to be in numeric format like 1, 2, 3 ...and till 31

Example
Use DATE function to calculate dates

Year	Month	Day	Result	Formula
2017	12	31	12/31/2017	=DATE(A22,B22,C22)
2016	11	30	11/30/2016	=DATE(A23,B23,C23)
2015	3	1	3/1/2015	=DATE(A24,B24,C24)
2014	8	1	8/1/2014	=DATE(A25,B25,C25)

Exercise 1
Use DATE function to calculate dates and then format the result into Date format

Year	Month	Day	Formula
2012	7	31	41121
2011	1	30	40573
2010	2	1	40210
2000	5	25	36671

Exercise 2
Use LEFT, MID and RIGHT functions to extract day, month and year from Col A and then use DATE function to calculate dates

Col A	Year	Month	Day	Date
21122016	2016	12	21	12/21/2016
13122000	2000	12	13	12/13/2000
15011999	1999	01	15	1/15/1999
10102010	2010	10	10	10/10/2010

Date - DATE | Date - DATEVALUE | Date - TODAY | Date - NOW | Date - DAY | Date - MONTH | Date - YEAR | Date - WEEKDAY

4. Time Function

MINUTE

What does it do?
This function returns the minutes between 0 and 59

Syntax
=MINUTE(SERIAL_NUMBER)

Serial Number
Date or time from which you want to extract minutes

Example
Identify today's weekday number basis different return_type value

Col A	Result	Formula
2/25/2026 21:01	1	=MINUTE(A15)
10:15 AM	15	=MINUTE(A16)
10:20 PM	20	=MINUTE(A17)

Exercise
Use MINUTE() to find out minutes from Col A

Col A	Results
1/31/2017 0:50	50
1/1/2017 0:01	1
12:45 AM	45

5. Lookup Function

Exercise 1
Below are 3 different tables, in the first table we have employees and their designation, in the second table we have their names and salary and in the 3rd one we have their names and department. Use Vlookup() to fill all these details in the 4th table

E Name	Designation
Ben	AVP
Ken	Analayst
Rachel	AVP
Katie	Analyst
Helen	Analyst

E Name	Salary
Ben	\$85,000
Ken	\$60,000
Rachel	\$65,000
Katie	\$50,000
Helen	\$45,000

E Name	Dept
Ben	Analytics
Ken	Analytics
Rachel	IT
Katie	HR
Helen	IT

E Name	Designation	Salary	Dept
Katie	Analyst	\$50,000	HR
Helen	Analyst	\$45,000	IT
Rachel	AVP	\$65,000	IT
Ben	AVP	\$85,000	Analytics
Richard	#N/A	#N/A	#N/A
Ken	Analayst	\$60,000	Analytics

Exercise 2
Below in the **Lookup** table, we have cost for different parts for different car makers. Use VLOOKUP() to get costs of items in the second table

Lookup Table	GearBox	Engine	Steering	Ignition	CYHead
Mercedes	\$2,500	\$5,000	\$1,250	\$250	\$300
BMW	\$2,000	\$4,999	\$1,302	\$300	\$290
Audi	\$3,000	\$5,200	\$1,100	\$200	\$310

Maker	Mercedes	BMW	Audi	Audi	Mercedes	Mercedes	BMW	BMW
Spare	CYHead	GearBox	Gearbox	Steering	Ignition	Engine	GearBox	Engine
Costs	\$300	\$2,000	\$3,000	\$1,100	\$250	\$5,000	\$2,000	\$4,999

6.Math Function

Exercise 2

Use IF and SUM function to calculate sum of the following values if conditions meet. If the conditions do not meet, then return value 0. First has been done as an example

Condition to meet	Col B	Col C	Col D	Col E	Result	
Col B > 1	5	1	2	4	12	=IF(B54>1,SUM(B54:E54),0)
Col C > 50	9	56	6	12	83	
Col C > 500	12	345	23	547	0	
Col E > 80	84.3	59.8	80.9	80.5	305.5	
Col D > 35	9.8	0.2	38	80.9	128.9	

Exercise 3

Use IF, AND and SUM functions to calculate sum of the following values if conditions meet. If the conditions do not meet, then return value 0. First has been done as an example

First Condition	Second Condition	Col B	Col C	Col D	Col E	Result	
Col B > 1	Col D >= 2	5	1	2	4	12	=IF(AND(C65>1,E65>=2),SUM(C65:F65),0)
Col C > 50	Col A > 5	9	56	6	12	missing COL A	
Col C > 500	Col E > 500	12	345	23	547	0	
Col E > 80	Col A > 80	84.3	59.8	80.9	80.5	missing COL A	
Col D > 35	Col B >= 9.8	9.8	0.2	38	80.9	128.9	

Exercise 4

Use IF, OR and SUM functions to calculate sum of the following values if either of the 2 conditions meet. If the conditions do not meet, then return value 0. First has been done as an example

First Condition	Second Condition	Col B	Col C	Col D	Col E	Result	
Col B > 1	Col D >= 2	5	1	2	4	12	=IF(OR(C76>1,E76>=2),SUM(C76:F76),0)
Col C > 50	Col A > 5	9	56	6	12	missing COL A	
Col C > 500	Col E > 500	12	345	23	547	927	
Col E > 80	Col A > 80	84.3	59.8	80.9	80.5	missing COL A	
Col D > 35	Col B >= 9.8	9.8	0.2	38	80.9	128.9	

7.Statistic Function

MIN

What does it do?

This function returns the smallest number in a given set of values. It ignores logical values and text

Syntax

=MIN(Number1,Number2 and so on...)

Number1,Number2 and so on...

List of values from which you want to find the minimum value

Example

Use MIN function to find the smallest value of the 3 numbers present for each row

Number1	Number2	Number3	Result	Formula
3	5	3	3	=MIN(A15:C15)
19	a	1	1	=MIN(A16:C16)
8	b	c	8	=MIN(A17:C17)
-8.5	3	0.5	-8.5	=MIN(A18:C18)

Exercise

Use MIN and IF function on the following and if the value is less than 5, then flag it as 'less than 5' else 'Greater than equal to 5'

Number1	Number2	Number3	Result
3	5	3	3
19	a	1	1
8	b	c	8
-8.5	3	0.5	-8.5